

# Data Structures

Course 2025-2026

## Theory and Lab lectures:

**Monday:** 12:00-14:00 → Theory → Classroom: A-020

**Thursday:** 12:00-14:00 → Lab → Classroom: A-331

## Lecturer:

Ugaitz Amozarrain

Department of Statistics, Computer Science and Mathematics

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## Office hours:

- Monday: 16:00-18:00
- Wednesday: 11:00-13:00

By default, otherwise stated, all office hours will be in my office, though it will be helpful if we agree on the time slot. You can contact me either through the internal email of MiAulario, or at ugaitz.amozarrain@unavarra.es. Thus, we coordinate the meetings and avoid crowds in the department corridor.

## Tentative Schedule:

| MONDAY                                 |    | THURSDAY                  |    |
|----------------------------------------|----|---------------------------|----|
| Jan 26th                               | 1  | Jan 29th                  | 2  |
| Presentation                           |    |                           |    |
| Feb 2nd                                | 3  | Feb 5th                   | 4  |
| Review Python basics / Tips and tricks |    | Sparse Matrices           |    |
| Feb 9th                                | 5  | Feb 12th                  | 6  |
| Object-Oriented Python                 |    | Working with Files        |    |
| Feb 16th                               | 7  | Feb 19th                  | 8  |
| Lists, Dictionaries, Stacks            |    | Postfix Calculator        |    |
| Feb 23rd                               | 9  | Feb 26th                  | 10 |
| Queues and Dequeues                    |    | Palindromes               |    |
| Mar 2nd                                | 11 | Mar 5th                   | 12 |
| Buffers                                |    | Balanced Parentheses      |    |
| Mar 9th                                | 13 | Mar 12th                  | 14 |
| Linked Lists I                         |    | Conway's Game of Life     |    |
| Mar 16th                               | 15 | Mar 19th                  |    |
| Linked Lists II                        |    | <b>Father's Day</b>       |    |
| Mar 23rd                               | 16 | Mar 26th                  | 17 |
| Linked Lists III                       |    | Working with Linked Lists |    |
| Mar 30th                               | 18 | Apr 2nd                   |    |
| Recursion and Computational Complexity |    | <b>Easter</b>             |    |
| Apr 6th                                |    | Apr 9th                   |    |
| <b>Easter</b>                          |    | <b>Easter</b>             |    |
| Apr 13th                               | 19 | Apr 16th                  | 20 |
| Binary Trees                           |    | Recursion Exercises       |    |
| Apr 20th                               |    | Apr 23rd                  | 21 |
| <b>Day of the University</b>           |    | Constructing Binary Trees |    |
| Apr 27th                               | 22 | Apr 30th                  | 23 |
| Generalizing Trees                     |    | Huffman Coding I          |    |
| May 4th                                | 24 | May 7th                   | 25 |
| Graphs                                 |    | Huffman Coding II         |    |

| MONDAY   |           | THURSDAY           |           |
|----------|-----------|--------------------|-----------|
| May 11th | <b>26</b> | May 14th           | <b>27</b> |
| Review   |           | Huffman Coding III |           |
| May 18th |           | May 21st           |           |
| May 25th |           | May 28th           |           |

### Exams:

**25/05/2026:** 16:00 → Final exam: Everything

**17/06/2026:** 16:00 → Retake exam: Everything

### Evaluation criteria:

| Evaluation method                                                                                                                                         | Weight ( %) | Retake policy |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------|
| Theoretical/Practical exams (Individually performed). Every student must get at least 50 % on the exams to average with the rest of the gradable aspects. | 50 %        | YES           |
| Deliverables                                                                                                                                              | 50 %        | YES           |

- Exam (5 points):
  - It is mandatory to pass the exam in order to pass the course.
  - The re-take exam is worth 100 % of the grade.
- Deliverables (5 points):
  - All deliverables will be checked to confirm correct delivery.
  - Each week you will have to deliver the code for the corresponding lab project.

| Lab Project               | Delivery/Correction | Max points                                                     |
|---------------------------|---------------------|----------------------------------------------------------------|
| Linked Lists              | Correction          | 2                                                              |
| Huffman Coding            | Correction          | 2                                                              |
| Sparse Matrices           | Delivery            | $\frac{n}{8}$ , where n is<br>the number of<br>delivered items |
| Working with Files        | Delivery            |                                                                |
| Postfix Calculator        | Delivery            |                                                                |
| Palindromes               | Delivery            |                                                                |
| Balanced Parentheses      | Delivery            |                                                                |
| Conway's game of Life     | Delivery            |                                                                |
| Recursion Exercises       | Delivery            |                                                                |
| Constructing Binary Trees | Delivery            |                                                                |